Anti-Forensic Challenges in Digital Forensics Investigations: An Overview of Techniques and Tools

Ibrahim Adam Mohammad
Center of Excellence in Cybercrimes and Digital Forensics (CoECDF)
Naif Arab University for Security Sciences,
Riyadh, Saudi Arabia
ibrahimbeem82@gmail.com

Amer Osman Nasar
Center of Excellence in Cybercrimes and Digital Forensics (CoECDF)
Naif Arab University for Security Sciences,
Riyadh, Saudi Arabia
amirzaheer99@gmail.com

Mohammed Alkhawaldeh
Center of Excellence in Cybercrimes and Digital Forensics (CoECDF)
Naif Arab University for Security Sciences,
Riyadh, Saudi Arabia
mohdabuyaman88@gmail.com

Emad-ul-Haq Qazi
Center of Excellence in Cybercrimes and Digital Forensics (CoECDF)
Naif Arab University for Security Sciences, Riyadh, Saudi Arabia.
qabdulrab@nauss.edu.sa

Tanveer Zia
School of Arts and Sciences
University of Notre Dame Australia
tanveer.zia@nd.edu.au

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***Abstract*—Techniques and tools have become an important part of forensic investigations, and evidence gathering in forensic cases. Most digital forensic experts use methods and tools during case investigations. At the same time, hackers and cybercriminals also use new techniques and tools called anti-forensics to distract, mislead, or prolong the investigation process. This paper provides a critical overview of anti-forensic techniques such as data hiding, trail obfuscation, artifact wiping, attacks against forensic tools, and the tools used in cybercrimes. The paper demonstrates two examples, one for data hiding because it is the most common type of anti-forensic technique and the other for altering file extension which is one of the easiest ways to mislead a forensic examination. Finally, the paper provides recommendations for digital forensics examiners on how best to handle anti-forensics challenges.**

Keywords— Anti-forensic, Anti-forensic techniques, Anti-forensic tools

1. Introduction

Technological advances have contributed to facilitating various services for people, institutions, and countries. However, the dependence on technology and the internet in various areas of life has increased the rate of cybercrime and the diversity of its methods. This can also be seen through the current situation in the COVID-19 pandemic, as during the period (January to April 2021) nearly 907,000 unwanted messages, 737 malware incidents, and 48,000 malicious URLs - all related to the pandemic were detected. An Interpol report on the assessment of cybercrime shows a significant increase in cybercrime during the COVID-19 pandemic [1] which includes attacks on health systems. Cybercriminals have exploited these circumstances to commit more cybercrimes such as email and internet fraud, cyber extortion, hacking, and cyber terrorism [2]. Therefore, it requires continuous updating, awareness, and development of investigators' tools for digital crimes.

Digital forensics is a field of forensic science that includes retrieving what is contained in digital devices and conducting examination and analysis to determine if the devices were used in a crime and assist in the process of proving or denying the crime [3]. One of the primary objectives of digital forensic investigation is to provide accurate, reliable, repeatable, and legally acceptable evidence capable of proving or denying cybercrime or contributing to the discovery of the perpetrators [4], and this is done by following the correct procedures and standards for digital forensic and maintaining the integrity of evidence, and the ability to illustrate techniques and procedures in digital forensics investigations.

Cybercriminals try to evade investigators and not leave traces of their crimes by concealing, destroying, modifying, and manipulating evidence. This is described as anti-forensics. There are four common anti-forensic techniques: data hiding, artifact wiping, trail obfuscation, and attacks against computer forensics [4]. In this paper, we will critically examine these techniques and the challenges that result from using these technologies that confront digital forensics processes. The paper also investigates tools used in these techniques.

Anti-forensic is defined in several ways in the literature.

- Definition (1) “Anti-forensic from the digital forensic perspective is the process of erasing or obfuscating digital forensic artifacts to avoid digital forensic investigations." [5].
- Definition (2) "Techniques used to prevent the application of digital forensics to those criminal and civil laws that are enforced by police agencies in a criminal justice system." [6].
- Definition (3) "The countermeasures taken to avoid forensic investigation and evade it. The major goal of the anti-forensic technique is preventing investigators from accessing evidence of their crimes" [7].

We define anti-forensic as any action taken by the user, using removal, modifying, and concealing digital evidence that makes the process of locating, acquiring, analyzing, and examining evidence more complex in a digital forensic investigation.

2. Anti-forensic Techniques

In this section, an overview of four commonly used anti-forensic techniques is provided.

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A. Data Hiding

As the term suggests, data hiding makes it difficult to locate data. Sometimes, it is referred to as a technique of security through mystery [5]. Data can be hidden using different techniques such as the following:

Steganography: A hidden message or words can be stored on a white background [8] or storing the message or text in audio files by replacing the data as confidential data using a specific encryption technique [7].

Data Encryption: Applied to protect access to stored data by encrypting files, emails, or disk portions using several algorithms [9]. Data encryption is used in cell phones, computers, and the transmission of data in networks.

Relocation of data: Data is stored at an unusual location to mislead the investigation. Another method for relocating data is to move data to a different roving storage device and delete it from the primary device [7].

Altering file extensions: Any files in the computer device are specified by their file extension. The file extension is appended to a file name [7].

B. Artifact Wiping

It is a process of deleting, destroying, or overwriting data or evidence which makes it difficult for the examiners to recover it. File Wiping: Completely erasing data from the storage media. This not just deletes but erases data in the disk sectors.

Disk/Partition Wiping: Applied to demolish or wipe data existing on all disk partitions.

Disk Degaussing is a process of displaying a storage device to a magnetic field to delete already stored data. Registry

Wiping: The registry is used to store settings and data for the operating system and applications. Registry wipers attempt to delete information constantly.

Metadata Manipulation: It is a process of manipulation of metadata of each file. Operating systems make metadata for every file [9].

C. Trail Obfuscation

It is the deployment of different tools and steps that muddies tracking of a computer crime. The major goal is to mislead or turn the investigator's line of investigation away from the offender's tracks [10]. Trail obfuscation is often achieved by erasing and/or modifying log files and/or system event files or modifying the timestamps of files [11]. There are two methods of trail obfuscation: Peer-to-peer networking: which is a common form of trail obfuscation; P2P software can be misused for criminal activity (e.g., copyright infringement, spread of malware, etc) [12]. IP address spoofing: this is a common technique used by cybercriminals where they modify the source address by changing their IP and MAC addresses [13].

D. Attacks Against Forensic Techniques

Cybercriminals have good technical skills; they use their capabilities to attack forensics tools. Techniques such as program packers, anti-reverse engineering, and attacks against the integrity of investigating parties are common tactics. Program packers are software tools that compress files and then unpack when a program is ready to be executed. Anti-reverse engineering needs to be considered in the classification of anti-forensics as criminals try to divert investigators to extract data from artifacts. However, program packers are technically a composite of anti-reverse engineering, the two were separately categorized; as more particular forms of anti-reverse engineering are identified, the classification will be updated to include them. A newer concept added to the classification was attacks against the integrity of forensic investigators, the offenders may try to extrude, frame, or perform other malicious activities to attack the integrity of the collective tasked with the legal investigation of computer criminal issues [11]. Table I. summarizes anti-forensic techniques and the main purposes of each one with examples.

TABLE I. THE MAIN PURPOSES OF ANTI-FORENSIC TECHNIQUES

Anti-forensic techniques	The main purposes
Data hiding	• Hide data in such a way that cannot be found by forensic detection tools. Example: hide data in image files.
Artifact wiping	• Wiping data to not be recovered by forensic recovery tools. Example: delete the operating system data stored in the registry using the registry wiping tool
Trail obfuscation	• Mislead the investigator with false data to make them search in the wrong place away from the perpetrator. Example: manipulate with the timestamp.
Attacks against forensic tools	• Attack the integrity of investigating parties and prolong the investigation time or analysis . Example: a zipped file with a massive amount of data when unzipped will waste the time of investigation.

3. ANTI-FORENSICS TOOLS

This section highlights widely used anti-forensic tools, which are software or techniques employed to wipe off digital evidence, making it challenging for forensic investigators to analyze and recover data. These tools often involve data encryption, file shredding, metadata removal, and other methods to hinder the forensic examination process.

A. Data Hiding Tools

Snow: This tool is used to hide messages in the text by inserting whitespace at the end of lines. Because spaces and tabs in most cases are not visible to the text viewers, the message is hidden from normal observers.

Xiao Steganography: This tool is a free tool that can be used to conceal secret documents in BMP images or in WAV files.

Steghide: This is a free steganography tool that allows users to hide secret text in pictures or sound files [15].

Bit-locker: This tool allows encryption of hard disk, devices, and encryption of Space.

VeraCrypt and Truecrypt: These tools are used to encrypt certain data or to create a virtual encrypted disk, that can be used wherever needed [9].

B. Artifact Wiping Tools

The following artifact wiping tools are reported by [13]: USB Oblivion: This tool is created to remove all traces of USB-connected disks and CD-ROMs from the Registry in Windows operating system.

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Eraser: This is an open-source tool for the Windows operating system that erases critical data from hard disk by overwriting it many times with cautiously selected patterns.

Disk Wipe: This tool is a free moveable Windows application for permanent disk data devastation. It can delete all data on a disk and prevent retrieval of those data.

C. *Trail Obfuscation Tools*

Timestamp is a commonly used tool that uses Metasploit Framework which is a free penetration testing framework to exploit systems on different platforms. After that, it modifies timestamps of files to mislead the user and investigator [13]. There are new versions of applications like BitTorrent or eMule that allow the users to encrypt the packets.

D. *Attacks Against Forensic Tools*

In the literature [14], the following tools are used to attack forensic tools:

Zip bombs: zip bombs are ordinary zip files; these files expand and generate a massive amount of data if when unzipped.

USBKill: This tool is helpful to protect USB ports, if an unknown USB is connected to the system, this tool can delete files on that USB. Figure 1 shows the four types of anti-forensic techniques and relevant tools that are discussed in this section. In Table II. anti-forensic techniques are provided in column 1 and relevant forensics tools that influence their performance are provided in the middle column with the features documented in the right column.

TABLE II. FEATURE OF ANTI-FORENSIC TOOLS

Anti-forensic Techniques	**Tools**	**Features**
Data Hiding	Snow	Uses white space at the end of the line to hide text in ASCII
	Xiao Steganography	Several encryption algorithms are used to mask files [15].
	Steghide	It prevents and protects the samples from testing the color frequency as it does not change the frequency of the color sample [15].
	VeraCrypt	Apply on-the-fly encryption (OTFE), apply either 200,000 or 327,661iterations by default (which can be modified according to the user to the extent of 2048), based on the algorithm used.[16]
	Truecrypt	Apply on-the-fly encryption (OTFE), uses 1,000 repetitions of the PBKDF2-RIPEMD160 algorithm for system blocks [17].
	Bit-locker	Apply the AES-256-CBC or AES-128-CBC that is an encryption algorithm in cipher block chaining with a (128) bit or (256) bit [18].
Trail obfuscation	Timestomp	Changes to a file's timestamps are undetectable by forensic investigators and file review software. It is built on the use of device features for malicious purposes [19].
	eMule	Transferring resources directly between clients, quickly recover corrupted downloads [20].
Artifact wiping	USB Oblivion	It allows connecting to devices by all USB connections.
	Eraser	Deleting data through the process of overwriting it more than once.
	Disk Wipe	• It can be used to run programs without having to install them • It has a limited file size • It can be played on both internal and external drives [21]
Attacks against forensic Tools	USBKill	• Can hide any traces of the program after closing it • Wipe Swap and Ram • Runs in sleep mode OS-X [21]
	Zip bombs	It nests files in the zip container without multiple copying of files. The input size of a zip bomb increases quadratically, which means that the pressure will increase if the bomb is enlarged [21].

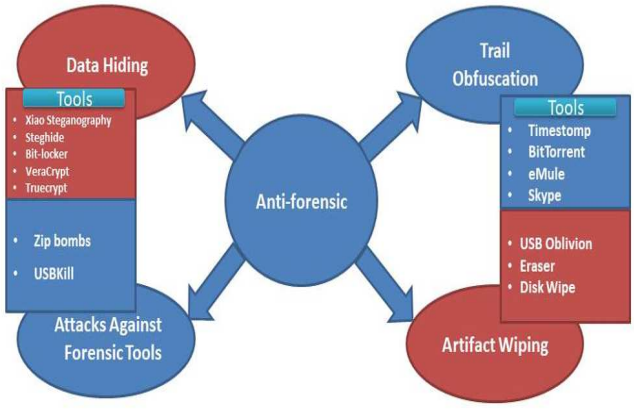

Figure 1: Anti-forensic techniques and tools

4. AN EXAMPLE OF A STEGANOGRAPHY TECHNIQUE

In this example, there are two text files, the first one named "nauss.txt" Figure 2 and the second one named "nauss_output.txt" Figure 3 and the same text has been written in both files as follows:

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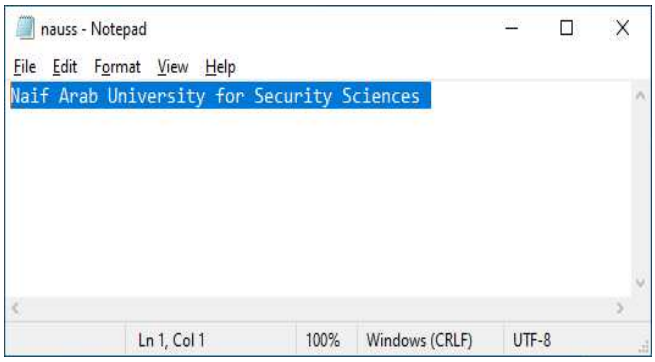

Figure 2: nauss.txt

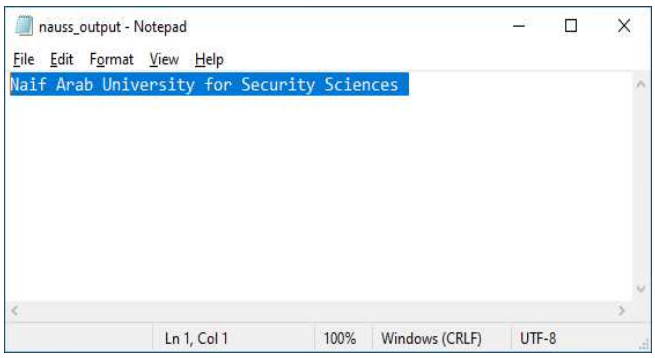

Figure 3: nauss_output.txt

Snow tool is used to hide data and lock it with password "123" using the following command Figure 4.

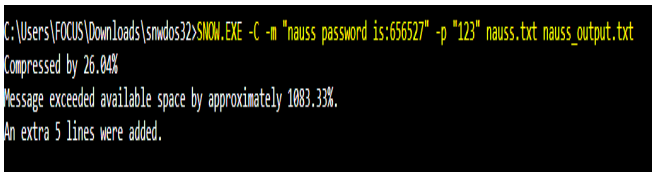

Figure 4: Snow tool command to hide data

Now see the difference before and after hiding data in the text file named 'nauss_output.txt' in Figure 5(a) and Figure 5(b).

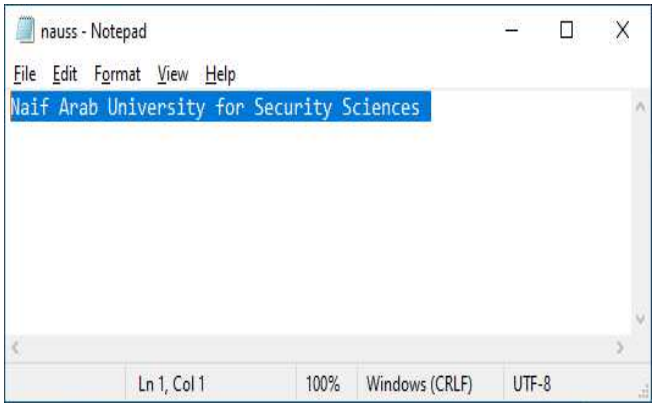

Figure 5(a): Text File without data hidden

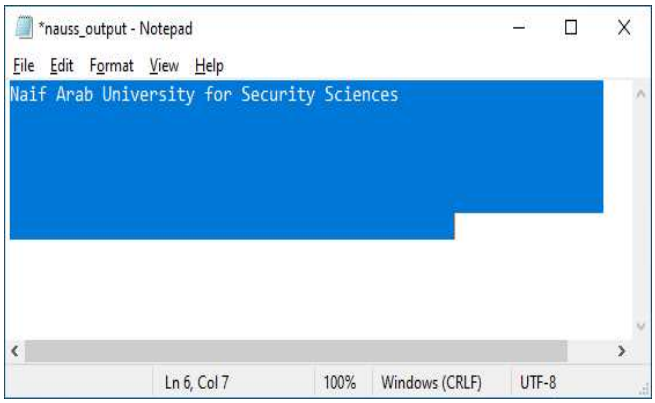

Figure 5(b). Text file with data hidden

Now we will show the hidden data using the password in Figure 6.

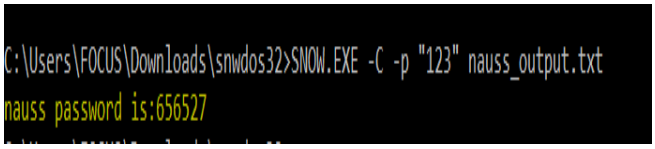

Figure 6: Revealing hidden text

5. An Example of Altering File Extensions Tehcnique

Local disk F": The file is located on the F drive, whereas Images is a directory on the F drive. It is a subfolder where the "NAUSS.jpg" file can be found. The file "NAUSS.jpg" is located on the F drive in a directory called "Images." The full path to this file would be something like "F:\Images\NAUSS.jpg.".

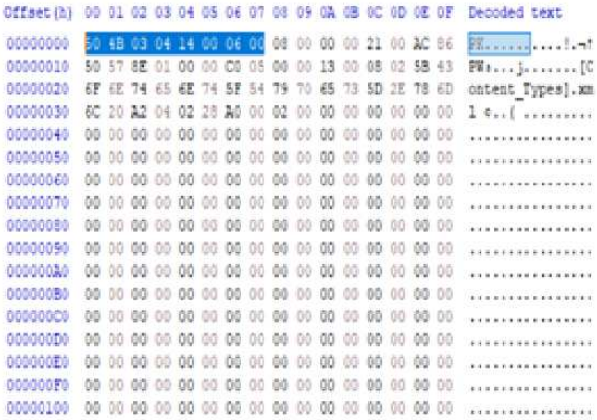

Figure 7: NAUSS.jpg

As we can see in Figure 7 the signature is 50 4B 03 04 14 00 06 00 which is the signature of a document, not an image file because the extension of the image file is FF D8 at the start and FF D9 at the end. Now we can alter the extension of file from .jpg to .docx to discover what it contains and open it in Microsoft Word.

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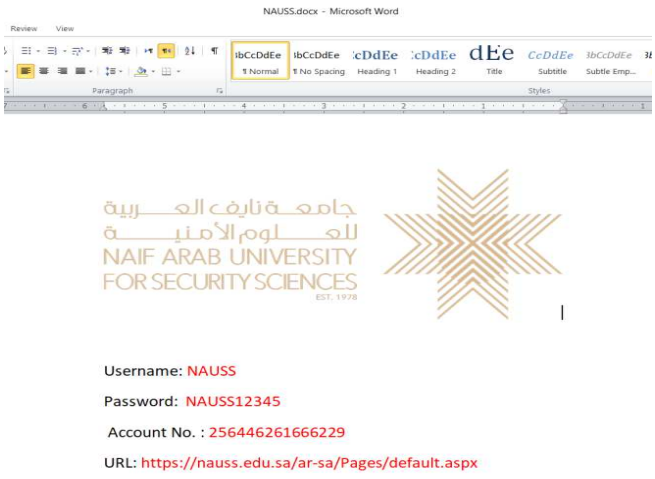

Figure 8: NAUSS.docx

Specifically, this file is said to contain sensitive information like usernames, passwords, account numbers, and a URL related to Naif Arab University for Security Science as shown in Figure 8.

6. Related Work

Anti-forensics is described by Jain, Anu, and Gurpal Singh Chhabra [5] as countermeasures used to thwart and evade forensic investigation. They investigated the efficacy of various anti-forensic tactics such as artifact wiping, data hiding, attacks against computer forensics tools (CFTs), and processes such as trail obfuscation to evade forensic investigation. In this study, two virtual machines and Access Data's FTK Imager were used for forensic research to demonstrate various anti-forensic techniques. Some of the techniques used include relocating data to an unexpected location, steganography, which is a technique for hiding information or files in other files and changing file extensions as examples of artifact wiping. They demonstrated various methods such as counterfeit evidence, defragmentation with the goal of confusing investigation, and modification of metadata, especially timestamps, using the Timestomp utility in trail obfuscation.

Computer forensics tools themselves are attacked where perpetrators use vulnerabilities or weaknesses to launch buffer overflow attacks and run arbitrary code on network forensic tools like tcpdump, snort, or ethereal, which could cause network forensic tools like tcpdump or snort to malfunction, as well as denial of service attacks [7].

Anti-forensics techniques (AFTs) and mitigation proposals against them were reviewed by Murat Gül and EminKugu [2]. They tested new and commonly used methods such as data pooling, non-standard RAID'ed disks, manipulating file signatures, restricted filenames, manipulating MACE times, loop references, hash collisions, dummy HDDs, and mitigation proposals against these anti-forensics' techniques. Volumes can be de-RAIDed on the suspect's device using a boot disk that uses their RAID controller to minimize this technique. Manipulation of File Signatures A file signature is a short block of hexadecimal code that is used to specify the file type. A hexadecimal editing tool can reveal the file signatures of a file [2].

Majed, Hussein, Hassan N. Noura, and Ali Chehab [7] defined the different forms of AFT based on the intent of use and the type of attacks performed on Artifact wiping, which is designed to fully delete and remove data and can be applied to files, entire disks, or partitions, and includes disk/partition wiping, disk degaussing, registry wiping, and metadata manipulation. Data hiding, which involves steganography, data contraception, file system manipulation, hard disk manipulation, which is network-based data hiding, is used to target the presence of data on storage that makes the study and inspection of digital evidence by forensics examiners difficult or impossible to perform. Trail obfuscation uses tactics like log manipulation, IP address spoofing, P2P networking, and proxy servers to confuse and disorient investigators. Program packers, attacks against the reputation of investigative parties, and AF that reduce footprint are examples of attacks against forensics resources and processes.

Program packers attack against the reputation of investigative parties, and AF that reduce footprint are examples of attacks against forensics resources and processes [9]. Evidence destruction, evidence source elimination, evidence hiding, and evidence counterfeiting are the four forms of anti-forensic identified by Harris [4]. The authors [4] also described a new form of anti-forensic called evidence source elimination, which was not mentioned in other studies. The idea behind this type is that no proof is ever made, so there is no need to destroy it [6]. Data contraception, also known as proof source avoidance, is the mechanism by which the attackers use tools and methods that are configured to leave no traces on the host operating system [5].

7. Discussion and Recommendations

To investigate any case where anti-forensic has been used, the forensic examiners should be familiar with anti-forensic techniques and tools that are used with each type of these techniques. The commonly used technique is to hide data where it is unlikely to be found. One of the most formidable techniques involves encryption, which is employed to render data unintelligible to unauthorized users [21]. Forensic examiners should be well-trained not only to identify these techniques but also to determine the tools used to hide the data.

Artifact wiping refers to wiping, deleting, or destroying data in such a way that cannot be recovered. Most of the wiping programs leave traces of wiping. Forensic examiners should be aware of the positions in which the data may be found after the process of deletion.

Trail obfuscation can mislead the investigation away from the offender's tracks. Examiners need to determine techniques very carefully before going deep in the investigation. Lastly, in attacks against forensic tools such as zip bombs that create a huge amount of data when they are unzipped, this type of technique needs extensive knowledge and expertise which makes the role of the examiners more challenging.

Here are some useful recommendations that examiners can use while working in the field of anti-forensics:

- Create and develop advanced tools, such as forensic Artificial Intelligence, to track and detect criminals and shorten the investigation time.

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- Have extensive training on the tools and techniques used by criminals to keep up with them and training in detection methods.
- Update forensic tools periodically and regularly.
- The examiners should handle the evidence carefully to be acceptable in court, for example, timestamps can mislead the investigation and change the sequences of events.
- Law enforcement agencies must have policies and guidelines that ensure how to deal with techniques and tools that govern the examiners to follow correct procedures.
- Developing Decoding and decryption techniques that make the examination much easier.
- Improving first responder procedures to make sure the process of maintaining devices in the power-on status helps to acquire RAM capture which may contain useful information such as recover passwords in plain text.

8. Conclusion

In this paper, we have reviewed four commonly used anti-forensics techniques that are attempts to hide data, wipe artifact, attacks on digital forensic methods and tools, and trail obfuscation that used to corrupt the evidence or mislead the investigation and hide traces left by cybercriminals. We have also described anti-forensic tools and have demonstrated two examples, one using the steganography technique to hide data and altering file extensions to mislead forensic investigations. We conclude that perpetrators use advanced techniques including encryption, data hiding, artifact wiping, and many other techniques to make the digital forensic examination more difficult. Anti-forensic requires more effort on the part of investigators and greatly affects the time taken to investigate and further skills in digital forensics. We have provided a set of recommendations that may help improve in detection of anti-forensic tactics.

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